4 A train has a maximum speed of 200 km / h. It accelerates from rest with constant acceleration of 0.70 m / s^2.

(a) (i) Define acceleration.

.....

..... [1]

(ii) Show that the maximum speed of the train is approximately 56 m / s.

[2]

(iii) Calculate the time taken for the train to reach its maximum speed.

time = ... [2]

(b) (i) The train has a total mass of 440 000 kg. Calculate the force which causes the acceleration of the train.

force = ... [2]

(ii) The train travels into a headwind. The force of this headwind opposes the motion of the train. State and explain the effect of this force on the motion of the train.

statement .....

explanation .....

.....

[1]

[Total: 8]

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